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Ecological Anthropology

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Concept of Ecological Anthropology

- Human adaptation is multi-dimensional and multi-hierarchical
- Humans can adapt biologically, culturally and socially
- Biologically, humans can adapt physiologically and genetically
- Hence, an holistic approach is needed that can describe such multi-faceted human adaptability – Ecological Anthropology

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Concept of Ecological Anthropology

- **It is a branch of anthropology which is concerned with bio-cultural responses and processes that affect the survival, reproduction, development, longevity and spatial distribution of human populations**
- Integrates findings from ecology, physiology, socio-cultural anthropology, geography etc
- Impetus for growth – failure of earlier approaches
  - Environmental Determinism
  - Natural Possibilism
  - Cultural Ecology

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Concept of Ecological Anthropology

- All these approaches failed to recognize the biological concept of ECOSYSTEM in which human relations are MUTUALISTIC
- Ecosystem concept emphasizes on SYSTEMS APPROACH
- Clifford Geertz – first anthropologist to use the concept of ecosystem in anthropology and Roy Rappaport and Andrew Vayda gave the strongest impetus to it

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Human Adaptation

What is Adaptation?

- The response of an organism to the structural and functional characteristics of its environment
- Adaptation happens through slow and gradual adjustments to the environment

Types of Adjustments – Rickelfs

- Regulatory Responses
- Acclimatory Responses
- Developmental Responses

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Types of Adjustments

1. **Regulatory Responses**
   - Ex. Cultural strategies
   - Psychological and Behavioral
   - Very rapid mode of adjustments

2. **Acclimatory Responses**
   - Take longer time
   - Changes in organic structure
   - Occurs when external stimuli are present for sufficient period of time
   - Usually reversible
   - Ex. Exercise – Muscles

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Types of Adjustments

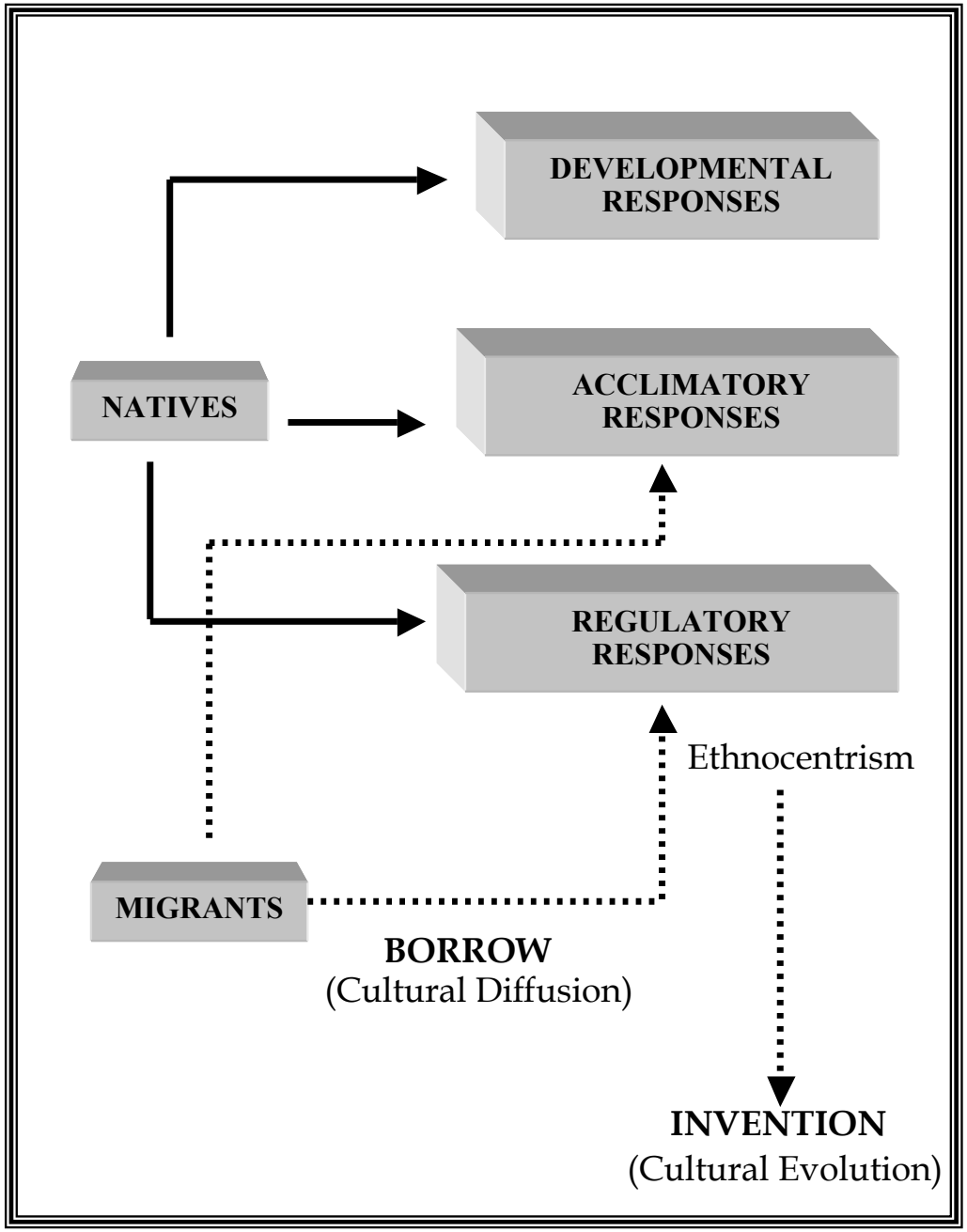
3. Developmental Responses

- Irreversible Responses
- Occurs during growth and development
- They are adjustments to environment present during developmental period
- Owing to greater plasticity of the individual during developmental period
- Ex. Child – High Altitude – Larger lung capacity

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The Multidimensional Nature of Human Adaptability



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Adaptation to High Altitudes

Stress – Hypoxia & Cold Stress

Adaptations to Hypoxia

- Increase in the number of capillaries
- Increase in pulmonary ventilation
- Larger lung and residual volume
- Polycythemia
- Work capacity and diet – high carbohydrate diet
- Reproduction and maturity
  - Low birth weight – reduction in oxygen demand
  - Delayed growth

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Adaptation to High Altitudes

Adaptations to Cold Stress

- Increased peripheral blood flow
- Warmer skin temperatures & non-shivering thermo genesis due to Brown Adipose Tissue (BAT)

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Adaptations to Arid Climate

Characteristics of Arid / Hot Desert Climate

- < 100 MM rainfall
- Dry and random distribution of rainfall

Stress

- Heat – Very high temperatures
- Water

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Adaptations to Arid Climate

- Diminished pulse rate
- Lower rectal temperatures
- Increased sweating
- Decrease in salt concentration of sweat
- Decrease in volume of urine
- Taller body sizes
- Slender bodes with no fat
- Aridity – Nomadic activity
- Clothing and Shelter

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